Causal Inference and Machine Learning for Strategic Policy Design in Health and Nutrition (Based on NFHS-5)

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Review of Previous Semester Work

1. Structural and Statistical Foundations of Causal Inference

- In-depth study of Completed Partially Directed Acyclic Graphs (CPDAGs) represent a Markov equivalence class of DAGs, identifying causal structures from observational data.
- Recursive structural decomposition methods for identifying causal order.
- Detailed analysis of the **PC Algorithm** and constraint-based learning.
- Integration of **False Discovery Rate (FDR)** control in sequential conditional independence testing.
- Study of **monotonic and signed causal effects** for structural validation.

Outcome: Built a rigorous theoretical foundation in causal structure learning, statistical reliability, and interpretability in high-dimensional settings.

Review of Previous Semester Work (Contd.)

2. Applied Causal Machine Learning Case Studies

Case Study I: Debiasing Recommender Systems

- Traditional collaborative filtering models implicitly assume that observed ratings reflect users' preferences; however, users' exposure to items is itself c**onfounded by platform policies and user behaviour**.
- Evaluated deconfounded estimators under **increasing bias**.
- Demonstrated improved robustness and interpretability.

Case Study II: Causal Drivers of Stock Drawdowns

- Estimated **causal effects of firm characteristics** during stress regimes.
- Implemented IPW, doubly robust methods, matching, and generalized random forests.
- Identified volatility (beta) as the strongest causal driver.
- Analyzed heterogeneous treatment effects across sectors.

Overall Progression: From theoretical DAG learning and statistical control to **real-world causal machine learning applications**.

The NFHS-5 Survey (National Family Health Survey)

NFHS-5 (2019–21) is India's most comprehensive health and demographic survey:

- $\sim$ 6.3 lakh households across all districts of India
- $\sim$ 7.2 lakh women and 1 lakh men interviewed
- Stratified two-stage sampling (rural–urban representation)
- Computer-Assisted Personal Interviewing (CAPI)

Why NFHS-5?

To move from abstract causal theory to real policy questions, we need a large, reliable, and structurally rich population dataset.

What makes it powerful for structural analysis?

- **Joint observation** of *socioeconomic status, health behavior, and biological outcomes*
- Objective biomarkers: height, weight, haemoglobin, blood pressure, glucose
- **District-level granularity**

Key Perspective: NFHS-5 is not just a dataset, it is a high-dimensional observational system where development indicators co-evolve.

From Association to Causality in Population Health

- Large population surveys reveal patterns. Policy, however, concerns **intervention**.
- Observational summary is given by: $E[Y \mid X = x]$, but we need causal question: $E[Y \mid do(X = x)]$

-In general,

$$E[Y \mid X = x] \neq E[Y \mid do(X = x)]$$

(Here, do-operator is an **intervention**, not observation, i.e. What happens to Y if we force X to be x).

Example:

- Does increasing female education improve child nutrition?
- Does sanitation expansion reduce anaemia?

-These require counterfactual reasoning: $\textbf{ATE} = E[Y(1) - Y(0)]$
-Population health indicators do not act independently. They arise from a structural system:

Policy design requires modelling the system, not just correlations.

- **Causal DAGs** encode these dependencies, clarifying adjustment sets and causal pathways.
- Machine learning then serves as a flexible estimator for high-dimensional components.

Causal Formulation of the NFHS Observational System

-With the causal framework in place, the next step is to embed it into a real population system.
-Instead of treating NFHS as isolated variables, we view it as a **structural system** where exposures, outcomes, and context interact.

- Y = health outcome (e.g., anemia, nutrition status)
- X = modifiable indicators (education, sanitation, etc.)
- Z = socioeconomic and environmental context

-**The policy question becomes**:

$$E[Y \mid do(X = x)] - E[Y \mid do(X = x')]$$

not simply $E[Y \mid X, Z]$, which reflects **association**.

-*Here,* $do(X = x)$ **represents an external intervention** *fixing* X*,* **independent of its natural causes (confounders)***, thus identifying true causal effect.*
-Causal Constructs and assumptions:

- Construct analytical dataset $\mathcal{D} = \{(Y_i, X_i, Z_i)\}_{i=1}^{n}$
- Ensure overlap: $P(X = x \mid Z = z) > 0$

Empirical Exploration: Distribution Diagnostics

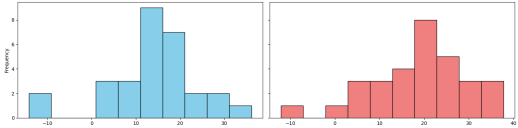
-Before structural learning, we examine the **statistical geometry** of the system.

-Observed behaviours:

1. Approximately symmetric indicator (**NMR**)
2. Left-skewed development indicator (**Institutional births**)
3. Right-skewed demographic variable (**Sex ratio at birth**)

-Variation across states is essential for meaningful **intervention contrasts.**

(1)



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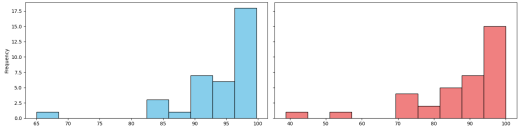

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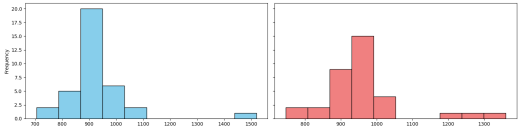

Pairwise Dependence Structure

-Pairwise correlation is computed using Pearson correlation:

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sqrt{\mathrm{Var}(X_i)\,\mathrm{Var}(X_j)}}$$

-The global dependence operator: $R = \{\rho_{ij}\}_{i,j=1}^{p}$

-High $|\rho_{ij}|$ values indicate latent coupling between indicators.

-From a structural perspective:

- Strong dependence may indicate shared **upstream causes.**
- **Correlation is not always causation.**

-**Important distinction**: $\rho_{XY} \neq$ causal effect

-Correlation maps geometry, not intervention.

Income-Stratified Correlation (SGDP-Based)

-Economic development acts as a **global structural confounder**.
-States **partitioned by per-capita SGDP**:

$$\mathcal{S} = \mathcal{S}_{low} \cup \mathcal{S}_{moderate} \cup \mathcal{S}_{high}$$

-**Mean stratified correlation**:

$$\bar{\rho}_{ij}^{SGDP} = \frac{\rho_{ij}^{(low)} + \rho_{ij}^{(moderate)} + \rho_{ij}^{(high)}}{3}$$

-Purpose:

- Suppress regime-specific noise
- Identify correlations stable across income gradients

Stable dependencies $\Rightarrow$ deeper structural relationships.

Area-Stratified Correlation (Rural–Urban)

-Area type (Rural or urban) introduces spatial heterogeneity.

$$R^{(a)} = \{\rho_{ij}^{(a)}\}$$

-Mean area-stratified correaltion:

$$\bar{\rho}_{ij}^{area} = \frac{\rho_{ij}^{(rural)} + \rho_{ij}^{(urban)}}{2}$$

-Interpretation:

- Persistent correlations across area types indicate structural coupling
- Instability suggests urbanization effects

-Multiscale dependence operators:

$$R, \quad \bar{R}^{SGDP}, \quad \bar{R}^{area}$$

-Together they map the dependency architecture of NFHS.

Correlation outputs

	Indicator_1	Indicator_2	Correlation	Abs_Correlation	Sign
83	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.955962	0.955962	Positive
188	Population living in households that use an im...	Women (age 15-49) who are literate4 (%)	0.754717	0.754717	Positive
103	Population below age 15 years (%)	Population living in households with electrici...	-0.746479	0.746479	Negative
117	Population below age 15 years (%)	Women age 15 years and above wih Elevated bloo...	-0.738019	0.738019	Negative
82	Female population age 6 years and above who ev...	Population living in households that use an im...	0.675057	0.675057	Positive
189	Population living in households that use an im...	Women (age 15-49) with 10 or more years of sc...	0.658316	0.658316	Positive

Figure: Pairwise Correlation sorted

	Indicator_1	Indicator_2	Mean_Correlation	Abs_Correlation	Sign
399	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.975438	0.975438	Positive
5436	Institutional births (in the 5 years before th...	Births attended by skilled health personnel (i...	0.959897	0.959897	Positive
3159	Neonatal mortality rate (per 1000 live births)	Infant mortality rate (per 1000 live births)	0.959770	0.959770	Positive
3262	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.942223	0.942223	Positive
5741	Births delivered by caesarean section (in the ...	Births in a public health facility that were d...	0.936711	0.936711	Positive
7241	Breastfeeding children age 6-23 months receivi...	Total children age 6-23 months receiving an ad...	0.936061	0.936061	Positive

Figure: Income stratified correlation

Conditional Dependence: Partial Correlation Analysis

-Raw correlations reflect co-movement, but development indicators often share upstream drivers. (e.g., income, literacy, socioeconomic status).

-To separate direct coupling from shared effects, we introduce **partial correlation**.
-For a single confounder Z:

$$\rho_{XY \cdot Z} = \frac{\rho_{XY} - \rho_{XZ}\rho_{YZ}}{\sqrt{(1-\rho_{XZ}^2)(1-\rho_{YZ}^2)}}$$

-More generally, via regression residualization:
$r_X = X - E[X \mid Z], \quad r_Y = Y - E[Y \mid Z], \quad \rho_{XY \cdot Z} = \mathrm{Corr}(r_X, r_Y)$
-Empirical findings:

- Conditioning on SGDP and literacy attenuated several strong raw correlations
- Some dependencies persisted $\Rightarrow$ potential structural linkage

-Partial correlation acts as a **statistical proxy for confounder** filtering before graph construction.

Correlation outputs (Continued)

	Indicator_1	Indicator_2	Mean_Correlation	Abs_Correlation	Sign
5436	Institutional births (in the 5 years before th...	Births attended by skilled health personnel (i...	0.970078	0.970078	Positive
7241	Breastfeeding children age 6-23 months receivi...	Total children age 6-23 months receiving an ad...	0.967085	0.967085	Positive
3262	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.957823	0.957823	Positive
399	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.938848	0.938848	Positive
5741	Births delivered by caesarean section (in the ...	Births in a public health facility that were d...	0.928641	0.928641	Positive

Figure: Area stratified correlation

	Indicator_1	Indicator_2	Partial_Correlation	Abs_Partial	Sign	Mean_Correlation_SGDP	Mean_Correlation_Area
7241	Breastfeeding children age 6-23 months receivi...	Total children age 6-23 months receiving an ad...	0.988377	0.988377	Positive	0.967085	0.936061
5436	Births attended by skilled health personnel (i...	Institutional births (in the 5 years before th...	0.984170	0.984170	Positive	0.970078	0.959897
3262	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.984027	0.984027	Positive	0.957823	0.942223
399	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.957087	0.957087	Positive	0.938848	0.975438
3159	Infant mortality rate (per 1000 live births)	Neonatal mortality rate (per 1000 live births)	0.946325	0.946325	Positive	0.823721	0.959770
5741	Births delivered by caesarean section (in the ...	Births in a public health facility that were d...	0.930874	0.930874	Positive	0.928641	0.936711
3160	Neonatal mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.922055	0.922055	Positive	0.758586	0.908272
5114	Children who received postnatal care from a do...	Mothers who received postnatal care from a doc...	0.912162	0.912162	Positive	0.914058	0.931261

Figure: Partial correlation

Proposed CPDAG and Mediation-Based Structural Refinement

-After filtering confounding via partial correlation, we proposed a causal DAG, encoding hypothesized causal pathways.
-Example structural chain:

$$X \to M \to Y$$

where X = exposure, M = mediator, Y = health outcome.

$$\textbf{Total Effect} = \text{Direct Effect} + \text{Indirect Effect}$$

-Using regression-based mediation:

$$M = \alpha_0 + \alpha_1 X + \epsilon_M \qquad Y = \beta_0 + \beta_1 X + \beta_2 M + \epsilon_Y$$

- Direct effect $\approx \beta_1$, and Indirect effect $\approx \alpha_1 \beta_2$
- Edges with **statistically negligible effects** or **statistically**

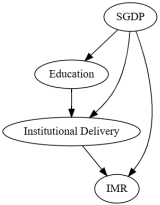

Figure: Proposed CPDAG

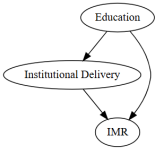

Figure: Corrected CPDAG

Hierarchical Clustering: Revealing Latent Development Blocks

-Pairwise correlation suggested organized co-movement.
-We convert similarity (pairwise correlation) into distance:

$$D_{ij} = 1 - |\rho_{ij}|$$

-Agglomerative clustering merges clusters by minimizing:

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} D_{ij}$$

-**Output**: a **dendrogram** revealing nested hierarchies.

- Education, digital access, institutional health cluster together
- Nutrition and maternal indicators form coherent submodules

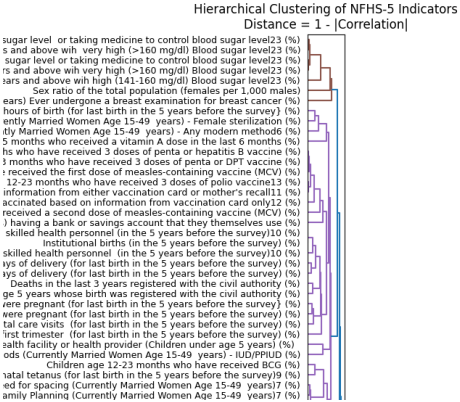

Figure: Hierarchical clustering (dendogram)

Graph Formulation and Louvain Community Detection

-Correlation matrix $\Rightarrow$ similarity network:

$$G = (V, E, W)$$

-Edges defined if $|\rho_{ij}| \geq \tau$, with weights: $w_{ij} = |\rho_{ij}|$
-Community detection via **modularity maximization:**

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

Louvain algorithm:

- Greedy modularity optimization
- Iterative aggregation of communities
- Fast and scalable

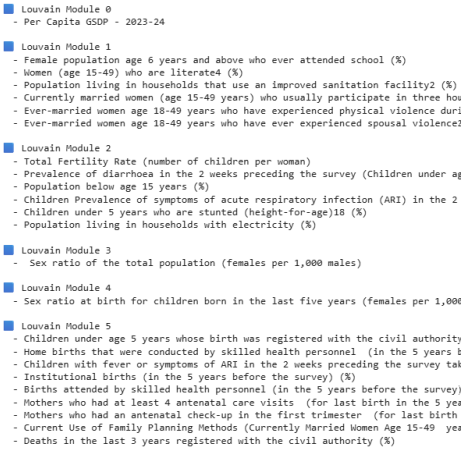

Figure: Louvain graph clustering

Leiden and Spectral Clustering: Robustness of Modules

Leiden **improves** Louvain by:

- Guaranteeing well-connected communities
- Refining modular partitions

Spectral clustering:

$$L = D - S$$

where L is the **graph Laplacian.**
-**Eigen-decomposition of** L embeds nodes in low-dimensional space, then clusters via k-means.
-Latent development modules are structurally stable.

Leiden Module 0
- Children under age 5 years whose birth was registered with the civil authority (%)
- Deaths in the last 3 years registered with the civil authority (%)
- Home births that were conducted by skilled health personnel (in the 5 years before
- Institutional births (in the 5 years before the survey) (%)
- Mothers who received postnatal care from a doctor/nurse/LHV/ANM/midwife/other health
- Births attended by skilled health personnel (in the 5 years before the survey)10 (%)
- Mothers who had an antenatal check-up in the first trimester (for last birth in the
- Current Use of Family Planning Methods (Currently Married Women Age 15-49 years) -
- Children with fever or symptoms of ARI in the 2 weeks preceding the survey taken to
- Children age 9-35 months who received a vitamin A dose in the last 6 months (%)

Leiden Module 1
- Mothers who consumed iron folic acid for 180 days or more when they were pregnant (f
- Mothers who consumed iron folic acid for 100 days or more when they were pregnant (f
- Births in a private health facility that were delivered by caesarean section (in the
- Men age 15 years and above wih high (141-160 mg/dl) Blood sugar level23 (%)

Spectral Cluster 0
- Health worker ever talked to female non-users about family planning (%)
- Current users ever told about side effects of current method of family planning8

Spectral Cluster 1
- Households using clean fuel for cooking3 (%)
- Population living in households with an improved drinking-water source1 (%)
- Men (age 15-49) who have ever used the internet (%)
- Children under 5 years who are stunted (height-for-age)18 (%)
- Women (age 15-49 years) who are overweight or obese (BMI â‰¥25.0 kg/m2)21 (%)
- Men (age 15-49 years) who are overweight or obese (BMI â‰¥25.0 kg/m2) (%)
- Women age 15-24 years who use hygienic methods of protection during their menstr
- Men (age 15-49 years) who know that consistent condom use can reduce the chance
- Women (age 15-49 years) who know that consistent condom use can reduce the chanc
- Women (age 30-49 years) Ever undergone a screening test for cervical cancer (%)

From Indicator Clusters to Latent Meta-Variables

-**To reduce dimensional complexity:**, for cluster C_k,

$$Z_k = \frac{1}{|C_k|} \sum_{X_i \in C_k} X_i$$

-Each Z_k represents a **latent meta-variable:**

$$\mathbf{Z} = \{\text{Education, Health , Infrastructure,}$$

Economy, Women Empowerment, Nutrition, Socioeconomic, Others $\}$

-Benefits:

- Cluster-level $\rightarrow$ Module-level structure.
- Enables higher-level causal reasoning

Causal Discovery via PC Algorithm

-Within and across clusters, we apply the PC algorithm.
-Core principle is independence testing:

$$X_i \perp X_j \mid S$$

-Edges removed if **conditional independence** holds.
-Orientation via collider rules: $X_i \to X_k \leftarrow X_j$
-**Output**: Complete Partially Directed Acyclic Graph or CPDAG (Markov equivalence class).
-**Adjacency encoding:** +1 if edge from i to j, -1 if from j to i, else 0.

-**Bootstrap aggregation of DAGs: Bootstrap confidence** of edges given by:

$$\hat{P}(i \to j) = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{i \to j \in G^{(b)}\}$$

-Only edges with **confidence** $> \tau$ (0.1 here) are retained.

Emergent Causal Pathways and Structural Insights

Two structural layers emerged:

Inter-Cluster DAG:

- Infrastructure → Nutrition
- Nutrition → Health
- Women Empowerment → Health

Intra-Cluster DAG (e.g.- Education, Health)

- Women's literacy → Improved sanitation use
- Women's education → Early marriage
- Women anaemia → Child anaemia
- Family planning → Child nutrition

Directed paths: $X_a \to X_{i_1} \to \cdots \to X_b$ correspond to mediation chains.

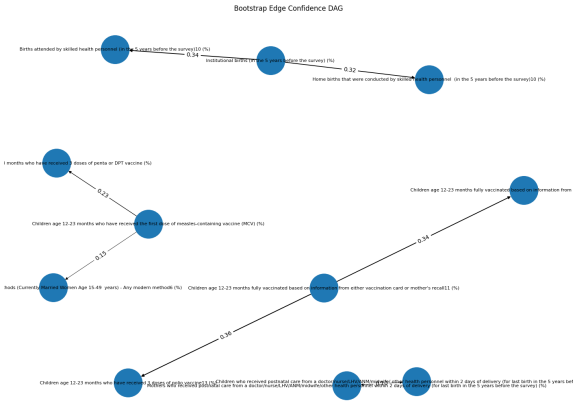

Figure: Intra-cluster edge confidence

Intra-cluster CPDAGs

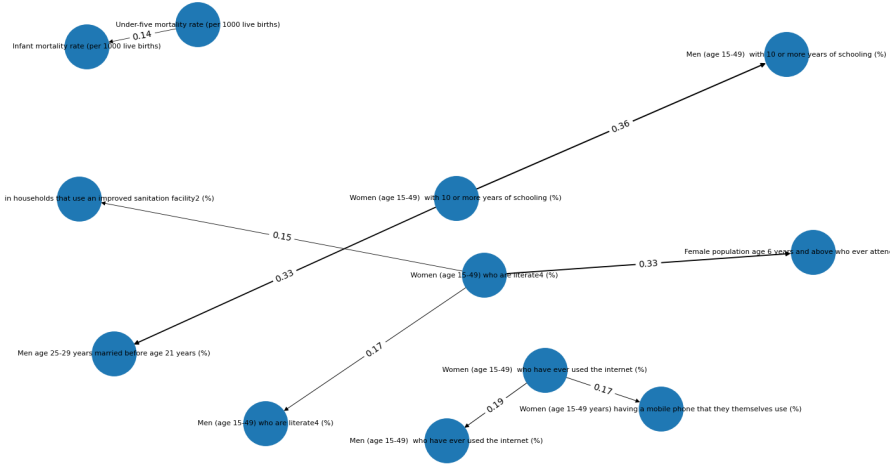

Figure: Intra-cluster edge confidence for cluster-2

Limitations and Future Research Directions

Current Limitations

- Cross-sectional observational data $\Rightarrow$ limited temporal causal validation
- Correlation-based clustering sensitive to threshold choice (τ)
- PC algorithm sensitive to sampling variability and CI test assumptions
- Possible false positive/negative edges despite bootstrap filtering

Next Phase: Structural Causal Learning Integration

- Use Louvain/Leiden/Spectral clusters to restrict PC conditioning sets
- Evaluate improvement in:
  - False discovery rate
  - Recovery of interpretable pathways
- Assess orientation accuracy, scalability, and noise robustness
- Operationalize identified pathways under potential outcomes framework
- Apply:
  - Propensity score matching
  - Inverse probability weighting (IPW)
  - Doubly robust estimators

Appendix (Code)

All **Python implementations,** experimental pipelines, and causal scripts are available in my **Google Colab notebooks** at the following links:

- **Notebook A: Exploratory Data Analysis and Distribution Diagnostics:**
  - **Google Colab Link:** `https://colab.research.google.com/drive/1emGnIb3MjfVTZj9Y_7bC_WE1QgOZo_8u`
- **Notebook B: Correlation Analysis, Graph Clustering, and Causal Discovery:**
  - **Google Colab Link:** `https://colab.research.google.com/drive/1NkJ3J4xEoT9i4J-h9LBUf-RYVyVFXG4d`

Thank You!

Questions and Discussion